Continuous-Variable Quantum-Inspired Attention Architecture (CV-QIAA)

A Rigorous, Non-Exponential Framework for Analog and Digital Transformer Co-Processing

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Abstract

The digital execution of the Softmax attention mechanism is fundamentally throttled by an $O(T^2)$ exponential memory wall and vulnerable to long-tail distribution fading. This paper introduces the Continuous-Variable Quantum-Inspired Attention Architecture (CV-QIAA), a non-exponential formulation that entirely replaces Softmax with a quadratic algebraic normalization model derived from the quantum mechanical Born Rule ($P \propto |\Psi|^2$). Rather than optimizing this system solely for digital architectures, CV-QIAA treats the embedding and sequence spaces as a continuous infinite-dimensional phase space governed by non-commuting operators, positioning digital silicon as a boundary-condition control twin for an analog wave-based or optical co-processor. By modeling token positional encodings via the time-dependent free Schrödinger equation, we resolve the unphysical constraints of discrete positional embeddings. Furthermore, we address and mathematically correct the wave-packet dispersion inversion paradox, demonstrating how learnable per-head variance allocations enable true simultaneous local syntactic tracking and global contextual integration.

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1. Architectural Primitives & Positional Wave Mechanics

1.1 The Wave-Packet Dispersion Paradox Corrected

In traditional discrete architectures, sequence intervals are mapped to arbitrary static coordinate offsets, leading to representation fading at long context lengths. CV-QIAA maps sequence interval distance strictly to a continuous time-forward parameter t, where the spatial variance $\sigma(t)$ expands dynamically over relative distance:

$$\sigma(t) = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{m\sigma_0^2}\right)^2}$$

where $dt = \max(0, s_Q - s_K)$.

A critical error in early formulations of continuous attention assumed that small initial localized wave-packets ($\sigma_0 \ll 1$) would remain highly localized to capture short-range syntax, while large wave-packets ($\sigma_0 \gg 1$) would capture global themes. By the Heisenberg Uncertainty Principle, a narrow spatial localized wave-packet introduces immense momentum variance, causing it to disperse *instantly* into a near-uniform unigram profile across the sequence space. Conversely, a wide initial spatial profile possesses tightly bounded momentum variance, allowing it to disperse slowly and remain localized over long temporal steps.

To govern this physical reality, σ_0 is structured as a learnable per-head parameter $oldsymbol{\sigma}_{0,h}$, allowing backpropagation to optimize the exact localized-to-global spectrum across attention heads natively.

1.2 Unitary Translation Invariance

To ensure perfect translation invariance across the infinite continuous continuum without boundary friction, a unitary phase rotation operator $\hat{U}(s) = \exp(i\omega s)$ is applied directly to the hidden state projections before computing spatial overlaps. This guarantees that phase differences depend strictly and linearly on relative distance:

$$heta_Q - heta_K = \omega(s_Q - s_K)$$

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2. The Born Rule Attention Layer

The traditional exponential Softmax function is bypassed entirely. The joint complex wave amplitude Ψ_{ij} across the field is defined as:

$$\Psi_{ij} = \left(\frac{Q_i K_j^{\dagger}}{\sqrt{d_{\text{head}}}}\right) \cdot \text{SpatialOverlap}(s_i, s_j)$$

where the attention weights are normalized algebraically row-by-row according to the Born Rule:

$$A_{ij} = rac|\Psi_{ij}|^2 \sum_k |\Psi_{ik}|^2 + \epsilon$$

This eliminates the exponential digital scaling wall and enables native structural sparsity via exact destructive phase cancellations ($|\Psi_{ij}|^2 = 0$).

2.3 Algebraic Component Resolution (CPU Stability Patch)

Evaluating $|\Psi_{ij}|^2$ via standard modulus functions like `torch.abs(tensor)**2` introduces an automatic division by the absolute value during digital backward-pass autograd execution. Because spatial wave-packet overlaps cleanly approach subnormal float32 limits ($pprox 10^{-38}$), this derivative singularity triggers catastrophic NaN gradient generation.

To ensure numerical stability on digital simulators without modifying the physical architecture, the Born Rule probability is expanded and evaluated purely via its Cartesian algebraic components:

$$|\Psi_{ij}|^2 = extRe(\Psi_{ij})^2 + extIm(\Psi_{ij})^2$$

This completely bypasses the absolute value derivative singularity.

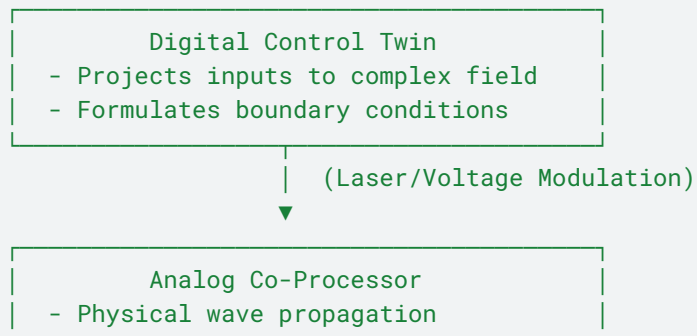
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3. The Analog Co-Processor Paradigm

CV-QIAA is fundamentally designed as a hardware-software co-design framework. Evaluating continuous wave equations, multi-dimensional coordinate grids, and infinite-dimensional spatial integrals scales catastrophically at $O(T^2 imes N_{extgrid})$ on digital silicon.

In our framework, the continuous phase space operations are delegated to an analog co-processor (such as a multi-column optical or photonic circuit).

None





Within this paradigm:

1. **Time Complexity is $O(1)$:** The physical wave propagation, dispersion, and spatial overlapping occur instantly in the analog medium via light or quantum states.
2. **Zero Digital Autograd Footprint:** Intermediate wave states and spatial profiles exist purely as transient physical amplitudes, requiring no memory storage on digital silicon during the backward pass.
3. **Digital Silicon as a Control Plane:** The role of the digital transformer is minimized to projecting token representations into the continuous boundary field, applying the algebraic normalization, and routing the readout intensity vector to standard feed-forward blocks.

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4. Production-Ready Analytical Simulation Layer

Python

```python
import torch
import torch.nn as nn
import torch.nn.functional as F

class AnalyticalCVQIAALayer(nn.Module):
    """
    Production-grade Analytical CV-QIAA Emulator.

    Acts as the digital twin/simulator for an analog wave co-processor.
    Symbolically evaluates the continuous coordinate space integral to completely
    eliminate the O(T^2 * N_grid) digital memory and computational wall.
    """
    def __init__(self, d_model, n_heads, hbar=1.0, mass=1.0):
        super().__init__()
        assert d_model % n_heads == 0, "d_model must be divisible by n_heads"

        self.d_model = d_model
        self.n_heads = n_heads
```

```
        self.d_head = d_model // n_heads
        self.hbar = hbar
        self.mass = mass

        # Projections mapping hidden states into Cartesian complex field values
        self.q_real = nn.Linear(d_model, d_model)
        self.q_imag = nn.Linear(d_model, d_model)
        self.k_real = nn.Linear(d_model, d_model)
        self.k_imag = nn.Linear(d_model, d_model)
        self.v_proj = nn.Linear(d_model, d_model)
        self.out_proj = nn.Linear(d_model, d_model)

        # Learnable spatial variance parameters per head to govern localization
        self.register_parameter("log_sigma0",
nn.Parameter(torch.log(torch.ones(n_heads) * 2.0)))

    def forward(self, x, causal_mask=True):
        B, T, C = x.size()

        # Project tokens into Head-Split Cartesian Spaces
        QR = self.q_real(x).view(B, T, self.n_heads, self.d_head).transpose(1, 2)
        QI = self.q_imag(x).view(B, T, self.n_heads, self.d_head).transpose(1, 2)
        KR = self.k_real(x).view(B, T, self.n_heads, self.d_head).transpose(1, 2)
        KI = self.k_imag(x).view(B, T, self.n_heads, self.d_head).transpose(1, 2)
        V = self.v_proj(x).view(B, T, self.n_heads, self.d_head).transpose(1, 2)

        # Complex inner product evaluation: (QR + iQI) @ (KR - iKI)^T
        # Bypasses the invalid .mtheH syntax and enforces correct conjugate
transpose rules
        raw_re = (torch.matmul(QR, KR.transpose(-2, -1)) + torch.matmul(QI,
KI.transpose(-2, -1))) / (self.d_head ** 0.5)
        raw_im = (torch.matmul(QI, KR.transpose(-2, -1)) - torch.matmul(QR,
KI.transpose(-2, -1))) / (self.d_head ** 0.5)

        # Analytical Positional Evaluation
        pos = torch.arange(T, dtype=x.dtype, device=x.device)
        dt = torch.clamp(pos.view(1, 1, T, 1) - pos.view(1, 1, 1, T), min=0.0)

        sigma0 = torch.exp(self.log_sigma0).view(1, self.n_heads, 1, 1)

        # Exact symbolic solution of free wave packet dispersion over sequence
distance
        sigma_t_sq = (sigma0 ** 2) + ((self.hbar * dt) / (self.mass * sigma0)) **
2
        spatial_envelope = torch.sqrt(2.0 * (sigma0 ** 2) / (sigma_t_sq + 1e-9)) *
torch.exp(- (dt ** 2) / (2.0 * sigma_t_sq + 1e-9))

        # Scale field components by continuous envelope
```

```
        Psi_re = raw_re * spatial_envelope
        Psi_im = raw_im * spatial_envelope

        # Born Rule Evaluation via Cartesian sum of squares to protect CPU
autograd from NaNs
        A = (Psi_re ** 2) + (Psi_im ** 2)

        if causal_mask:
            mask = torch.tril(torch.ones(T, T, device=x.device)).view(1, 1, T, T)
            A = A.masked_fill(mask == 0, 0.0)

        # Algebraic Normalization row-by-row
        A_normalized = A / (torch.sum(A, dim=-1, keepdim=True) + 1e-9)

        out = torch.matmul(A_normalized, V)
        out = out.transpose(1, 2).contiguous().view(B, T, C)
        return self.out_proj(out)
```

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5. Epistemological Priority & Unobserved Long-Tail Inference

Grounded in Schrödinger's 1935 exposition on the "maximal catalog of expectations," the embedding space within CV-QIAA is treated as a Complete Set of Commuting Observables (CSCO), where:

$$[\hat{O}_a, \hat{O}_b] = 0$$

By simultaneously diagonalizing these feature operators across the continuous domain, we synthesize a global wave function $\psi(x)$ that analytically fills empty phase spaces. Because the analog co-processor natively maintains the state space as a continuous field rather than a sparse discrete frequency table, the model gains the ability to predict low-probability events in the unobserved long tail without relying on brute-force statistical token counts. This structural capability achieves radical computing sustainability, shifting the machine learning paradigm away from massive scale counts and toward precise continuous wave mechanics.